

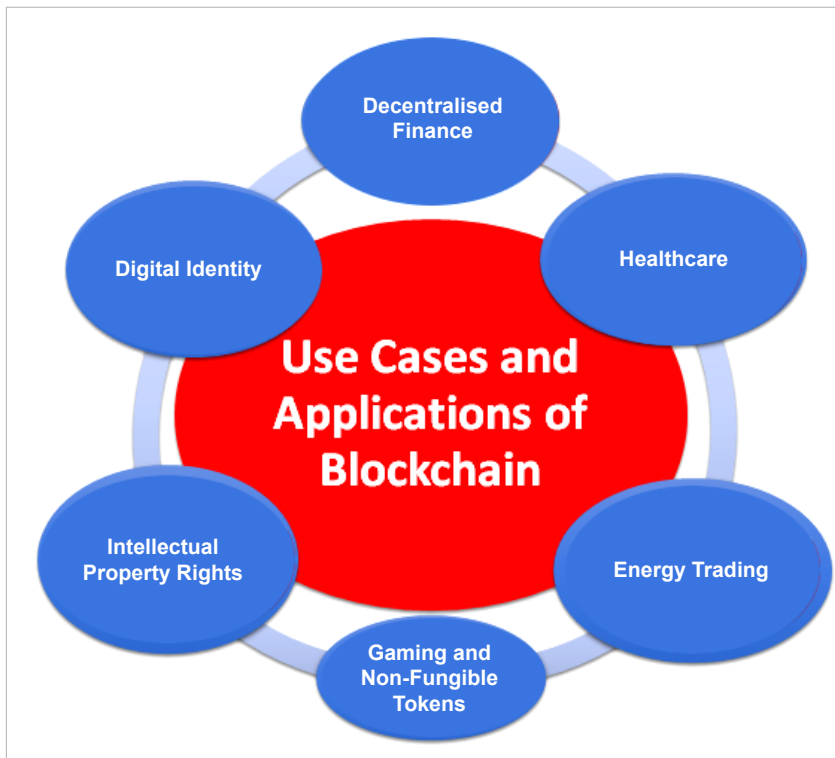


Figure 1: Key applications and use cases of blockchain

protocols. These applications enable users to engage in financial activities without intermediaries, enhancing transparency, reducing costs, and offering a wide range of financial services.

- **Healthcare:** Blockchain can securely store and share patients' medical records across healthcare providers. This ensures data accuracy, privacy, and reduces administrative inefficiencies. Patients have more control over their data, granting access only to authorised parties.
- **Voting systems:** Blockchain-based voting systems offer secure and transparent elections. Each vote is recorded as a transaction, preventing tampering and ensuring the integrity of the electoral process. This can help enhance voter trust and reduce instances of fraud. Decentralised voting systems use smart contracts to conduct transparent and tamper-resistant elections.
- **Digital identity:** Blockchain can create self-sovereign identities that users control. This could replace traditional identification methods, allowing individuals to manage and share their personal information securely while reducing identity theft and data breaches.
- **Energy trading:** Blockchain enables peer-to-peer energy trading, where individuals with solar panels can sell excess energy to their neighbours. This decentralised approach can make energy distribution more efficient and environment-friendly.
- **Intellectual property rights:** Blockchain can be used to prove ownership and protect intellectual property rights. Digital assets like artwork, music, and patents can be securely registered and tracked on the blockchain, preventing unauthorised use.
- **Cross-border payments:** Blockchain-based remittance services offer faster, cheaper,

and more secure cross-border transactions. Traditional banking intermediaries are bypassed, reducing fees and delays. Smart contracts simplify cross-border transactions by automating currency exchange and payment execution. This eliminates intermediaries, reduces fees, and accelerates the transfer process.

- **Gaming and non-fungible tokens:** The ability of blockchain to prove ownership has led to the rise of blockchain-based gaming and digital collectibles. Players can truly own in-game assets, and unique digital items can be bought, sold, and traded securely. Smart contracts are at the heart of blockchain-based gaming and NFT platforms to enable the creation, ownership, and trading of unique digital assets.

Smart contracts and decentralised applications

Smart contracts have emerged as a cornerstone of innovation within decentralised applications (DApps). The integration of smart contracts brings a multitude of benefits to various sectors, revolutionising how decentralised applications operate and interact with users.

Smart contracts are self-executing contracts with the terms of the agreement directly written into code. They automatically execute when predefined conditions are met, reducing the need for intermediaries and streamlining processes in areas like real estate, insurance, and legal agreements.

While smart contracts offer numerous advantages, they also come with challenges, including coding vulnerabilities and legal considerations. As the technology evolves, careful planning, code auditing, and integration within a comprehensive DApp ecosystem are vital for harnessing the full potential of smart contracts.